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Roll No

MI-7003-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mine Environmental Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. What are the various precautions for prevention of spontaneous heating of coal in mines? Discuss in detail. 14
2. a) What are the preventive measures of underground fires? 7
b) What are various methods of reopening mines? 7
3. a) Describe Dry Powder Extinguishers and CO₂ Extinguisher with neat sketch. 8
b) Explain rescue and recovery work in connection with mine fire. 6
4. a) Discuss the pressure balancing arrangements attached with sealed of area. 5
b) Describe Fire Damp Explosion and Coal Dust Explosion along with its preventive measures. 9

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5. a) Describe various types of stone dust barrier along with its applications 7
b) Discuss the provisions for establishment and location of "Rescue Station" and "Rescue Room" as per MRR. 7
6. a) Discuss the precautions to be taken while approaching old workings. 3
b) Define: Firedamp, White damp, Stink damp Blackdamp. 4
c) List out different types of rescue equipment used in mines. Explain any two with neat sketches. 7
7. a) Define a Disaster? What are the different causes of disasters in Indian coal mining industry? Explain any inundation disaster which occurred in Indian coal mining industry. 8
b) Write note on design and construction of Water Dams. 6
8. Write short notes on (any four): 14
 - a) Personal Protective Equipment
 - b) First Aid Station
 - c) Methane Layering
 - d) Water barrier
 - e) Soda Acid Fire Extinguishers.

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